TT-P-97d
January 25, 1968
SUPERSEDING
Int. Fed. Spec. TT-P-0097c (COM-NBS)
September 16, 1966 and
Fed. Spec. TT-P-97b
April 16, 1965

FEDERAL SPECIFICATION

PAINT, STYRENE-BUTADIENE SOLVENT TYPE, WHITE (FOR EXTERIOR MASONRY)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a white styrene-butadiene paint for application to exterior masonry surfaces.

1.2 Classification. Paint shall be of but one type, a white self-cleaning ready-mixed masonry paint, as specified hereinafter.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.

TT-P-442 - Pigment, Titanium Dioxide, (For Protective Coatings).

SS-S-346 - Siding, (Shingles, clapboard, and sheets) Asbestos-Cement.

Federal Standard:

Fed. Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-S-15191-Silica, Diatomaceous (Flatting-Extender Pigment).

MIL-Z-15486-Zinc Oxide, Technical (Acicular, Paint Use).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8010

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3. REQUIREMENTS

3.1 Material. Materials shall be as specified herein. Material not definitely specified shall be of a good commercial quality entirely suitable for the purpose intended under normal conditions of use. The paint shall be free from material which will be toxic to personnel under normal conditions of use.

3.2 Composition. The composition of the paint shall be as specified herein.

3.2.1 The quantitative requirements of the paint shall be as specified in table I. The requirements for pigment composition shall be as specified in table II.

3.3 Qualitative requirements.

3.3.1 Condition in container. The paint as received shall be well ground and not caked, livered, thickened, skinned or badly settled in the container. Any settling shall be capable of being easily broken up and redispersed in the liquid portion to form a smooth paint of good brushing consistency when tested as specified in 4.4.1.

3.3.2 Color. The color shall be white and shall match a reference standard if so specified by the procuring agency.

3.3.3 Brushing, flowing, and leveling. The paint shall show easy brushing, good flowing, and good leveling properties. The paint shall not pull, sag, or have a quick set under the brush when tested as specified in 4.4.1 at a spreading rate of 400 square feet per gallon.

3.3.4 Compatibility with thinner. There shall be no evidence of incompatibility of any of the ingredients of the paint when tested as specified in 4.4.2.

3.3.5 Self-lifting. The surface shall produce no lifting or other film irregularities after recoating when tested as specified in 4.4.1 and 4.4.4.

3.3.6 Flexibility. The paint shall show no cracking when subjected to the flexibility test specified in 4.4.5.

3.3.7 Skinning. The paint shall not skin within 48 hours in a 3/4 full container when tested as specified in 4.4.1.

3.3.8 Odor. The odor of the paint immediately upon opening the container and during application shall be normal for the product. (See suggested formula, table IV)

3.3.9 Accelerated weathering resistance. When the paint is tested as specified in paragraph 4.4.1 and 4.4.8, the paint film shall show no more than very slight chalking and no other film defect. The film shall not darken more than two units as determined from the daylight apparent reflectance; e.g., paint with an initial reflectance of 80 percent shall not drop below 78 percent. The chalk rating shall not be less than 8.

TABLE I. Quantitative requirements

Characteristics	Requirements	
	Min.	Max.
Pigment (percent by wt. of paint)	43.5	48.5
Vehicle (percent by wt. of paint)	51.5	56.5
Nonvolatile vehicle (percent by wt. of vehicle)	22.5	27.5
Consistency K U at 77°F.	65	78
Absorption (inch)	--	1/4
Drying time:		
Set to touch (hr.)	1/4	3/4
Dry hard (hr.)	--	2
Fineness of grind	3	--
Reflectance	80	
85 - Degree specular gloss (Sheen)	--	5
Weight per gallon (lbs.)	10.8	11.2
High styrene 1/-butadiene resin (percent by wt. of nonvolatile vehicle)	50	--

1/ 83-89 percent bound styrene

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TABLE II. Pigmentation

Pigments	Requirements	
	Min.	Max.
Titanium dioxide TT-P-442, type III (percent by wt. of pigment)	33.0	37.0
Zinc oxide MIL-Z-15486 (percent by wt. of pigment)	9.5	10.5
Extender 1/ (percent by wt. of pigment)	52.5	57.5

1/ The extenders used shall be silica, diatomaceous, MIL-S-15191 and talc (Al & Mg Silicate).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be performed in accordance with section 1000 of Fed. Test Method Std. No. 141 as applicable.

4.3 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined to determine compliance with section 5 of this specification.

4.4 Test procedures

4.4.1 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141, except as otherwise specified herein. The test methods are shown in table III.

TABLE III. Index

Characteristics	Requirement reference	Applicable test	
		Fed. Test Method Std. No. 141	Paragraph reference
Condition in container	3.3.1	3011	--
Consistency	Table I	4281	--
Brushing, flowing, and leveling	3.3.3	4321	--
Compatibility with thinner	3.3.4	--	4.4.2
Pigment content	Table I	4022	--
Vehicle (total vehicle in paint)	Table I	4052	--
Nonvolatile (in vehicle)	Table I	4041	--
Absorption	Table I	4421	--
Drying time	Table I	4061	--
Fineness of grind	Table I	4411	--
Reflectance	Table I	6121	--
85 - Degree specular gloss (Sheen)	Table I	6103	--
Self - lifting	3.3.5	6252	4.4.4
Flexibility	3.3.6	6221	4.4.5
Skinning	3.3.7	3021	--
Odor	3.3.8	4401	4.4.3
Accelerated weathering resistance	3.3.9	6152	4.4.8
Weight per gallon	Table I	4184	--
Pigments:			
Titanium dioxide	Table II	7081	--
Zinc oxide	Table II	7091	--
Extenders	Table II	7261	--
Vehicle:			
Styrene-butadiene determination	Table I	--	4.4.6
Resin identification	Table I	--	4.4.7

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4.4.2 Compatibility with thinner*. Pour 80 milliliters (ml.) of paint into a 100-ml. graduated cylinder and gradually stir in a thinner composed of equal parts by volume of an aromatic and alphatic blend of solvents having a Kauri-Butanol value of 60 or higher, (the aromatic portion must be of a lower volatility than the aliphati portion). Continue adding and stirring until the 100-ml. graduation is reached. Settling of pigment is disregarded in determining homogeneity.

4.4.3 Odor. Observe the odor immediately upon opening the container and during application for compliance as specified in 3.3.8.

4.4.4 Self lifting. Brush a coat (method 2141) of the material on a glass panel (method 2021) of approximately 1 mil dry film thickness and allow to dry for 24 hours at room temperature. Recoat the panel using the same method and after the film has dried for 24 hours examine for compliance as specified in 3.3.5.

4.4.5 Flexibility. Apply a film of paint with a Bird film applicator or equal to a wet film thickness of 0.002 inch on a tin panel (method 2012). Air dry for 18 hours, bake for 3 hours at 105° + 2°C. Cool for 1/2 hour and bend over a 1/8 inch mandrel. Examine for compliance as specified in 3.3.6.

4.4.6 Determination of styrene-butadiene resin.

Apparatus: Supercentrifuge
Centrifuge
Vacuum oven
Double beam grating infrared recording spectrophotometer
(spectral resolution 3 cm or better over entire range at slow scan)
Rock salt plates for use in spectrophotometer

4.4.6.1 Determination of total extractables (styrene-butadiene resin plus chlorinated paraffins 40 percent and 70 percent (1:1 ratio).

Weight 50 to 55g. of well-mixed sample to the nearest mg. using a weighing bottle and wash with benzene into a supercentrifuge bowl of 300 ml. capacity. Mix well and dilute with benzene to approximately 80% of capacity of the bowl. Benzene may be easily lost during centrifugation. Make sure the gasket is in good condition and the cover is sealed tightly. Centrifuge for 1/2 hr. at 50,000 rpm.

Cool the bowl externally before opening. Decant the liquid into a tared 250 ml. beaker containing three Berl saddles (as boiling stones) and evaporate to dryness on a steam bath. Extract the material remaining in the bowl by thoroughly mixing with benzene and proceeding as before. Decant into a second tared 250 ml. beaker containing three Berl saddles and evaporate to dryness. Generally the first benzene extract is free of pigment and the second extract has a small amount. The presence of pigment in the decantate is indicated by a cloudy benzene solution.

When pigment is present in either extract, remove it by the following procedure. Evaporate the benzene extract to dryness on the steam bath, take up in benzene and transfer to two 100 ml. conical-shaped centrifuge tubes. Centrifuge at 2000 rpm. for 45 minutes and decant into the original beaker and again evaporate on the steam bath. If both extracts contain pigment, centrifuge in order, using the same centrifuge tubes to eliminate mechanical loss.

Dry beakers in a regular oven at 70°to 75°C. over night. Then dry in a vacuum oven at 60°to 65°C and 30" of mercury to constant weight (approximately 32 hrs.). Weigh beakers and calculate total weight of extractables.

When the total weight of the extracts is obtained, dissolve both in benzene and combine. Keep the benzene solution as concentrated as possible. From this benzene solution, make films on sodium chloride plates using a medicine dropper. Air dry films to remove excess benzene. Select 6 to 7 films of a thickness such that the absorbance maximum of the bands at 3.3 microns and 3.5 microns lies between 0.5 and 0.7. Vacuum dry the films over night or 16 hours at 60°to 65°C and 30" of mercury. Scan the spectrum of each film from 2.5 to 15 microns using air as a reference and a slow scanning speed. Draw a baseline from 3.0 to 4.0 microns as illustrated in figure 1. Calculate the 3.3 μ / 3.5 μ ratio and from an average of 6 to 7 ratios, obtain the percent styrene-butadiene resin from a standard curve.

* Styrene-butadiene is not soluble in highly polar solvents such as acetone and alcohol.

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TABLE IV. Suggested formula (controlled chalking)

	Pounds per 100 gallons
Styrene-butadiene resin 1/	73.5
Thickener 2/	
Titanium dioxide, rutile	177.4
Zinc oxide	50.5
Asbestine	204.1
Diatomaceous silica	75.9
Chlorinated paraffin, 40 percent	35.3
Chlorinated paraffin, 70 percent	35.3
Aromatic hydrocarbon solvent 3/	225.9
Mineral spirits	225.9
	1103.8
Solids percent 59.1	
Percent PVC 57.1	

1/ Such as Pliolite S-5A or Marbon 9200

2/ Thixcin-R or Bentone 38 at 4-5 pounds per 100 gallons can be used to adjust viscosity; a gel produced by 10 percent Pliolite S-3 resin in a 50:50 mixture of aliphatic and aromatic hydrocarbons is also useful.

3/ Such as Solvesso 100.

4.4.6.2 Preparation of standard curve. The standard curve relates the 3.3 u / 3.5 u absorbance ratios to the percent styrene-butadiene resin. Prepare mixtures of high styrene-butadiene resin (such as Pliolite S-5A or Marbon 9200) and chlorinated paraffins (such as Chlorowax 40 and Chlorowax 70) with the percent of total weight for styrene-butadiene at 10.00, 25.00, 50.00, 75.00, and 100. Since it is difficult to weigh chlorinated paraffin 40 percent to the nearest 0.1 mg., it is best to weigh this first in a weighing bottle, taking an arbitrary weight. Rinse into a 150 ml. beaker with benzene and add an exactly equal amount of chlorinated paraffin 70 percent. When this is dissolved, add a proportionate amount of styrene-butadiene resin to yield the desired percentage. Make certain that all materials are completely dissolved in benzene. Produce a series of films for each formulation as described in the preceding paragraph but with the absorbance maximum of the strongest band between 0.5 and 0.7. Dry them in the same manner and calculate the 3.30 u / 3.50 u ratios from their infrared spectrograms. Plot the average of the absorbance ratios for each formulation vs. the percent styrene-butadiene resin using the large graph paper. The plot is a curve and not a straight line.

4.4.6.3 Calculation: Total wt. of extractables x percent styrene-butadiene resin = total wt. of styrene-butadiene resin

$$\frac{\text{total wt. of styrene-butadiene resin} \times 100}{\text{wt. of paint sample}} = \text{percent styrene-butadiene resin in paint sample}$$

4.4.7 Identification of resin from vehicle. Compare infrared spectrogram of the sample with the spectrogram of the known 50 percent styrene-butadiene resin mixture used in the preparation of the standard curve. (see figure 2). These styrene-butadiene paint resins are of the high styrene type. Draw the baselines as illustrated in figures 1 and 2 and determine the 10.3 u / 3.28 u ratios and compare. The ratio ranges from 0.95 for 83 percent styrene to 0.65 for 89 percent styrene. The ratio should be in this range.

4.4.8 Accelerated weathering resistance. Apply the paint by brushing at a rate of 400 square feet per gallon to the smooth side and edges of two asbestos-cement panels conforming to SS-S-346. Apply two coats allowing 24 hours drying between coats and 72 hours before testing. The first coat shall be reduced 8 to 1 part by volume or 1 pint thinner to 1 gallon of paint, the second coat shall be applied without reduction. Determine the daylight apparent reflectance as specified in 4.4.1 before and after 300 hours of accelerated weathering. After the accelerated weathering, examine the panels for chalking and other film defects and determine chalk rating by method 6411.

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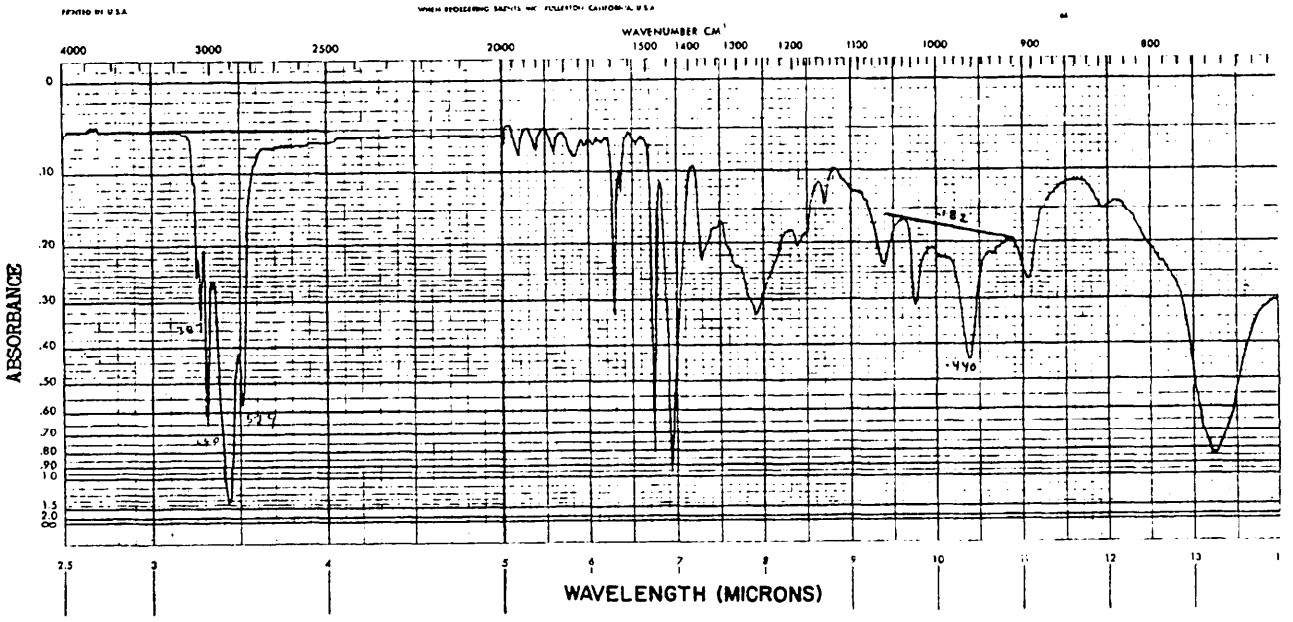


FIGURE 1. Extract of actual paint sample based on formulation in table IV.

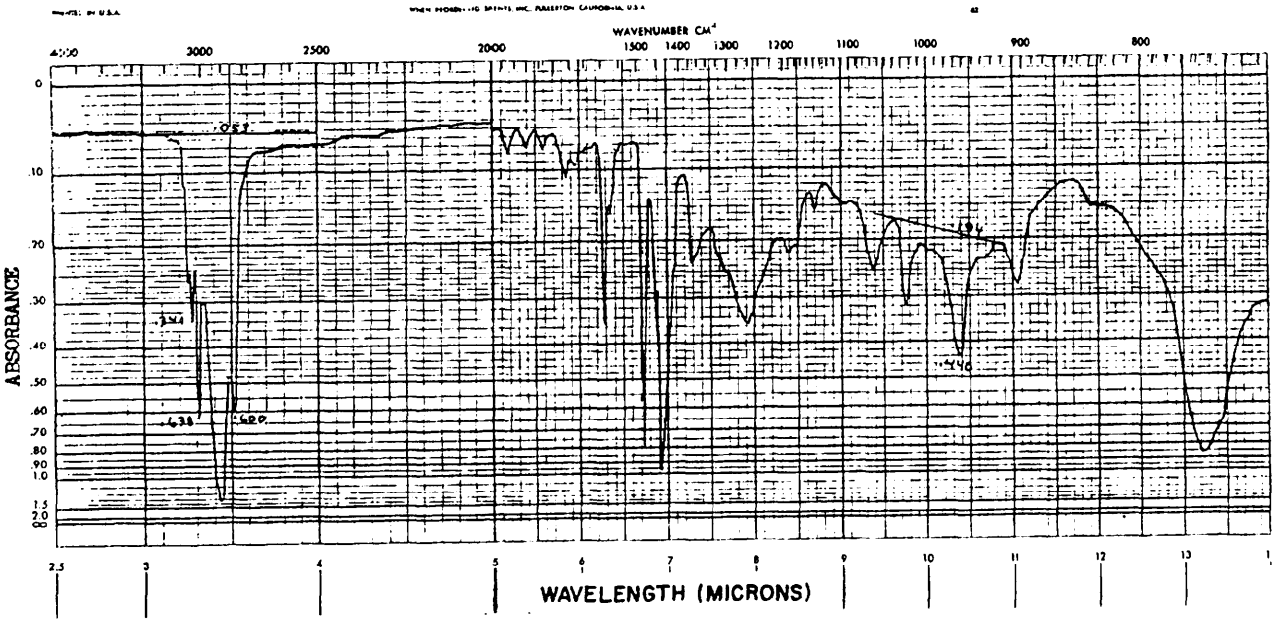


FIGURE 2. Mixture of styrene-butadiene resin, chlorinated paraffin 40 percent and chlorinated paraffin 70 percent in the proportion 50.00: 25.00: 25.00.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be A, B, or C and the level of packing shall be A, B, or C as specified (6.2). The paint shall be in 1-gallon metal cans, or 5-gallon steel pails as specified (6.2).

6. NOTES

6.1 Intended use. This specification covers white styrene-butadiene type paint for application to exterior concrete, stucco, and masonry surfaces which have been properly prepared. Surface preparation: on new work, no unusual preparation is required except in the case of new concrete, cinder block, or other extremely porous surfaces which should be filled or grouted. If filling or grouting is necessary, the product covered by TT-P-1098 is recommended. On previously painted surfaces, all loose powdery or flaking material, heavy chalking, and dirt should be removed by sand blasting, wire brushing, or scraping. It is further recommended that a two-coat system be used with the first coat, reduced 8 to 1 part by volume or 1 pint thinner (4.4.2) to 1 gallon of paint, and the second coat applied without reduction. This paint is intended for use as a white paint only. It is not suitable for use as a tint base. When colored paint is required, use styrene-acrylate type paint covered by TT-P-1181.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Administrative provisions for inspection records. (see 4.1).
(c) Level of packaging and packing required (see 5.1).
(d) Size of unit container (see 5.1).

6.3 Basis of purchase. The paint should be purchased by volume, the unit being a gallon of 231 cubic inches at 15.6°C. (60°F).

Preparing activity:

COM-NBS

CIVIL AGENCIES INTEREST:

GSA-FSS
HUD-HAA

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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